MIL-S-37845(DLA-DM)
17 October 1977
SUPERSEDING
MIL-S-37056
17 May 1974

MILITARY SPECIFICATION

SYRINGE, IRRIGATING, EAR AND ULCER, PLASTIC, DISPOSABLE

This specification is approved for use by the Defense Personnel Support Center, Defense Logistics Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a sterile, disposable, precision-molded vinyl ear syringe.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following specifications and standards of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

MILITARY

MIL-S-36586 - Sterilization Test Strip Set, Bacterial Spore.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from procuring activitiy or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D-2240 - Indentation Hardness of Rubber and Plastic by Means of a Durometer.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa 19103.)

UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (USP).

(Applications for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The ear syringe components (molded parts) shall be fabricated from medical grade polyvinylchloride (PVC) and shall be stabile, compatible and non-toxic in accordance with the requirements specified herein.

3.1.2 Compatibility. The materials of which the syringe is manufactured shall not react with, or be acted upon, by the liquids with which it will be in contact during normal use and operation, nor shall there be any materials incorporated in the syringe which will cause any deleterious effect when the syringe is used for the purpose intended.

3.1.3 Toxicity. The PVC shall meet the requirements for Class II plastics as stated in the Pharmacopeia of the United States (USP) when tested as specified in 4.4.1.

3.1.4 Heavy metal content. The plastic material shall contain not more than 5 parts per million (ppm) of lead when tested as specified in 4.4.2 or alternatively, 4.4.2.1. Heavy metal content may be determined by Atomic Absorption via Analysis, however in all reference cases, the lead method as described in the USP shall be used.

3.1.5 Durometer hardness. The Durometer hardness shall be A 53 $\pm$ 2 Shore when tested as specified in 4.4.3.

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3.2 Style and design. The ear syringe shall be of two piece vinyl construction, bulb style design, welded to form one integral unit. The bulb shall have molded sure-grip rings to provide positive control. The molded tip shall be free of any restrictions or occlusions and shall be properly tapered. It shall have a nominal capacity of 3 ounces (oz) and a minimum suction power of 3 inches (in) Hg.

3.2.1 Configuration and dimensions. The configuration and dimensions shall be in accordance with figure 1.

3.2.2 Size identification. The syringe shall be color coded for easy, positive size identification. Unless otherwise specified, the 3 oz syringe shall be green in color.

3.3 Sterility. The syringes shall be sterile, when tested with a biological indicator, as specified in 4.4.4.

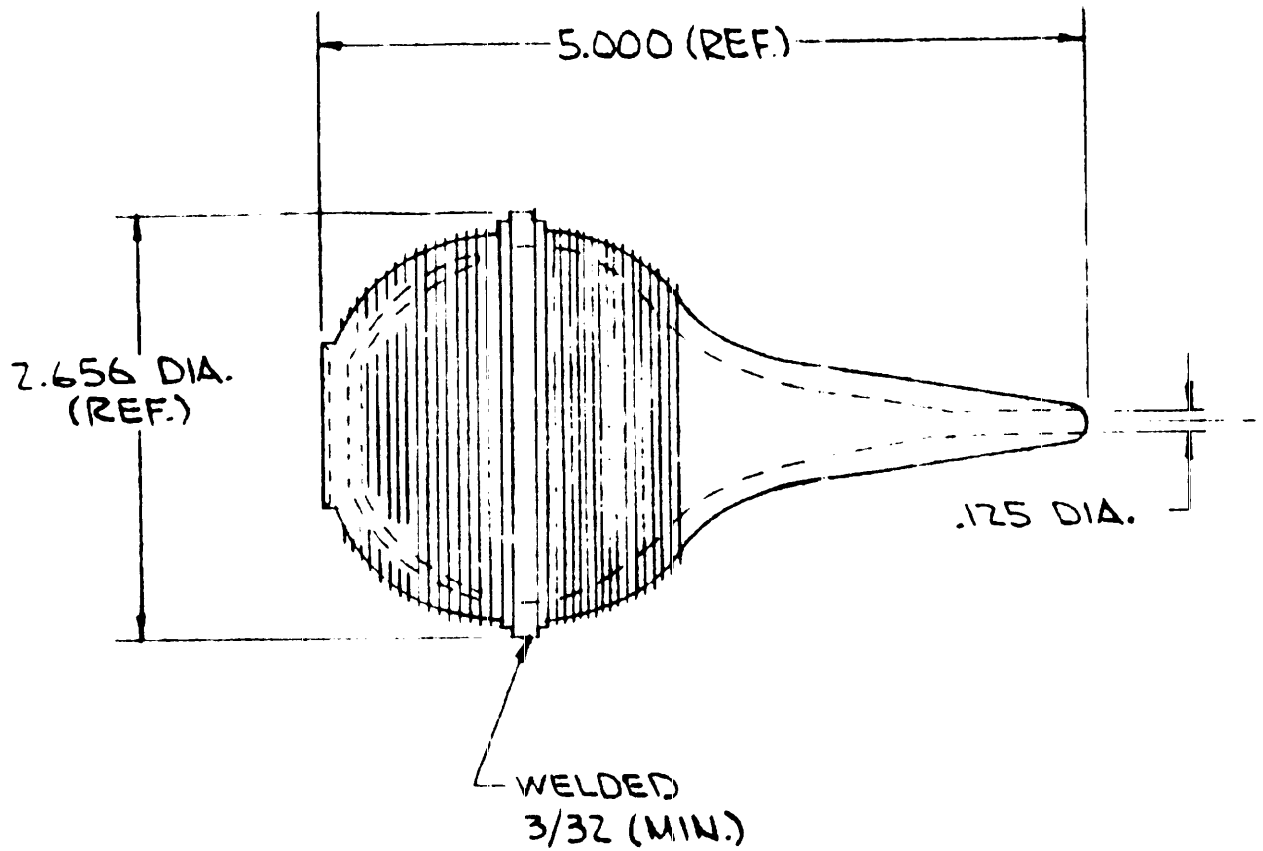
3.3.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National stock number (NSN).
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.3.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and sterilized simultaneously in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.2. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

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NOTE:
REMOVE ALL
FLASH.

TOLERANCES:
FRACTIONS ____ ± 1/64
DECIMALS ___ .XX ±.01
.XXX ±.005

FIGURE 1. Syringe, irrigating, ear and ulcer, plastic, disposable.

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3.4 Manufacturer's record. The manufacturer shall place a code or lot number on the individual package which can be verified against the manufacturer's records of all sterility and biological tests performed on the syringes. These records shall be made available for inspection by any duly authorized Government representative. No shipment shall be made until all tests have been completed, verified, and recorded in the permanent records of the manufacturer.

3.5 Storage life. When properly packaged and packed, the syringes shall be capable of being stored for a period of five years at a temperature of 59 - 86° F, without showing any evidence of deterioration such as cracks, blisters, softening or brittleness that would lessen the resiliency and useful life of the item or in any way impair its serviceability. Upon removal from storage, the device shall meet all the requirements set forth herein.

3.6 Storage life identification. Devices furnished under this specification shall not be marked on the device itself. Identification shall be made by marking the packing container or other suitable marking method acceptable to the procuring activity. The date shall be the date of manufacture of the device and shall be indicated by quarter of year (example: 3Q77 indicates a manufacturing date of calendar year 1977).

3.7 Workmanship. The finished syringes shall be symmetrical in shape, shall have smoothly finished surfaces, and shall be free from irregularities or imperfections of any kind which detract from their appearance or which impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the plastic, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of product for examination purposes shall be one syringe.

TABLE I. Sampling for examination purposes.

Type of defect	Inspection level	AQL (percent defective)
Major	I	1.0
Minor	S-2	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purposes shall be one syringe. Where feasible, the same sample shall be used for the determination of two or more tests characteristics.

TABLE II. Sampling for tests.

Unit of product	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Material	Toxicity	3.1.3	4.4.1.	2/	-
Class II	Metal content	3.1.4	4.4.2		
Plastic					
	Hardness	3.1.5	4.4.3	S-2	4.0
Syringe	Sterility	3.3	4.4.4	S-2	1/

1/ Acceptance number shall be zero for all sample sizes.
2/ Sufficient samples to perform the biological tests shall be selected from each lot.

4.2.3 Raw material. Ten grams of each lot of plastic used in the fabrication of the syringe shall be selected for the lead test.

4.3 Examination.

4.3.1 Examination. The syringe shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination shall not be restricted to defects listed in table III. Any syringe in the sample containing one or more defects shall be rejected and if the number of defective syringes in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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TABLE III. Classification of defects.

Categories	Defects
Critical	None defined.
Major	
101	Design not as specified.
102	Device not soft, pliable and resiliant.
103	Orifice not open or blocked by foreign material.
104	Device not free from bleeding, blooming or tackiness.
105	Orifice end not smooth and free of flash.
106	Color not as specified.
Minor	
201	Dimensions and configuration not as specified.
202	Device not free of pits, blisters, scratches, or crazing.
206	Flash not removed.

4.3.2 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests.

4.4.1 Biological tests. The syringes shall be tested in accordance with the requirements of the USP for class II plastic.

4.4.1.1 Animal implantation test. Samples of the syringes shall be tested in accordance with the method given in the USP for Implantation Test for class II plastic, except for the changes which follows: Last paragraph, second line, after "72 hours" add "and not to exceed 144 hours". Last paragraph, third and fourth sentences, delete entirely and substitute: "The tissues surrounding the implants shall be examined, both grossly and microscopically, by a pathologist certified by the American Board of Pathology or the American College of Veterinary Pathologists."

4.4.2 Test for leac. Ten grams of plastic shall be taken, ignited to char, and then further ignited in a muffler at 700° C to a white ash. The ash shall be boiled in 15 milliliters (ml) of 10 percent of volume of concentrated nitric acid (specific gravity of 1.4, chemically pure or analytical reagent grade), filtered into a volumetric flask and made up to 100 ml. An aliquot portion, equivalent to one gram of sample, shall be taken and placed in a separator. The dithizone test for lead, as given in the current edition of the USP, shall be applied using a control of 5 ppm or 12 ppm of lead solution as given in the USP test. If the sample solution is darker than the limit, smaller aliquots shall be used until the color of the control is matched by the sample solution and the concentration of lead in ppm is determined.

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4.4.2.1 Alternate test for lead. Accurately weigh a 5-gram sample into a 1-oz. Vycor crucible. Reduce to ash in a muffle furnace at 650° C for eight hours. Let cool to room temperature and add 15 ml of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil for 2 to 3 minutes, remove, let cool, and filter through Whatman #2 filter paper into a 50-ml volumetric flask. Rinse the residue on the filter with deionized water and dilute to 50 ml. Prepare standard lead solutions of 1, 2 and 5 ppm lead. Determine the absorbances of these solutions at the 283 millimicron wave length using a Hollow Cathode Pb lamp and a 10x scale expansion on a spectrophotometer, such as the Perkin-Elmer Model 303 Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

Calculation:

$$\frac{A \times F \text{ dilution}}{W} = \text{ppm lead}$$

A = Absorbance

F = Factor = $\frac{\text{(Concentration Standard in ppm Pb)}}{\text{absorbance}}$ 1/

W = Sample weight in grams

1/ The three standards are averaged to give ppm/unit A.

4.4.3 Durometer hardness. Durometer hardness shall be determined in accordance with ASTM D-2240. Hardness readings shall be made on test specimens fabricated from the same material batch used in the manufacture of the syringe.

4.4.4 Sterility test. The syringes shall be tested for sterility in accordance with the method specified in the USP for non-parenteral devices using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates it from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in 4.4.4.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in table III.

4.4.4.1 Inoculated product or simulated product. For each type of sterilization the organism used shall be as listed below:

Type of Sterilization	Organism	Temperature
Steam	B. stearothermophilus	55° - 65° C
Dry Heat	B. stearothermophilus, or	55° - 65° C
	B. subtilis var. niger (or globigii)	30° - 35° C
Gas	B. subtilis var. niger (or globigii)	30° - 35° C
Radiation	B. pumilus	30° - 35° C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

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4.4.4.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.4 and 4.4.4.1. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.4.1.
NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.4.4.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Request for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.4.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the USP.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing fifty individually packaged syringes constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Individual package. Each syringe shall be packaged in a sealed peel-open type container capable of maintaining sterility of contents unless the container is damaged or opened. The container shall be designed with tabs to facilitate opening.

5.1.2.2 Unit package. Fifty individually packaged syringes shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Fifty syringes individually packaged as specified in 5.1.2.1, shall be packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A, B or C, as specified (see 6.2).

5.2.1 Level A. Four unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Four unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged syringes shall be packed in shipping containers in a manner that will afford adequate protection against damage during shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

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5.3 Marking.

5.3.1 Individual package. In addition to any commercial markings, each package shall be marked to identify contents, and shall include the lot or control number and the legend or similar statement: "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED", and "STERILE".

5.3.2 Unit packages. Each unit package shall be marked as specified in MIL-STD-129. The method of sterilization (G- as, R-Radiation, S-Steam, C-Cold or D-Dry Heat Sterilization) shall be shown following the level of packaging and date of packaging. Example: "A 2/77-G". The sterility lot or control number shall also be shown. Final sterilization shall be on or after indicated packaging date. Marking shall include the legend "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

5.3.3 Shipping container. Shipping containers shall be marked as specified in MIL-STD-129. The method of sterilization as specified in 5.3.2 and the sterility lot or control number shall also be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

6. NOTES

6.1 Intended use. The bulb type syringe is a single use sterile device for aspirating mucous from the nose and pharynx in resuscitation of newborn infants.

6.2 Ordering data. Purchasers should exercise any desired options specified herein, and the procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Applicable levels of packaging and packing, (see 5.1 and 5.2).
(c) NSN.

6.3 National stock number coverage. This specification is intended to cover the following item appearing in the Federal Supply Catalog:

NSN	Item Identification
6515-01-045-0029	SYRINGE, IRRIGATING, EAR AND ULCER, 3 oz, Disposable, 50s

Preparing activity:
DLA-DM

Project No. 6515-2113

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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